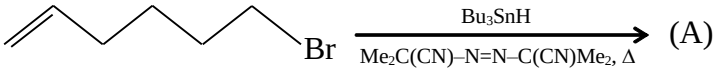
CHEMISTRY

SECTION – I

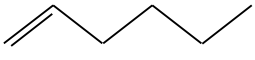
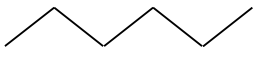
Single Correct Choice Type

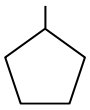
This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. 

The major product (A) of the given reaction would be

(a)  (b) 

(c)  (d) 

2. The rate of a certain reaction depends on concentration according to the equation:

$$-\frac{d[A]}{dt} = \frac{k_1[A]}{1+k_2[A]} \quad \text{where } k_1 > k_2$$

Which of the following statement is not correct for the above reaction?

(a) At low concentration of A, ($[A] << k_2$) the order of reaction is unity.

(b) At high concentration of A ($[A] >> k_2$), order of reaction is two.

(c) At low concentration of A ($[A] << k_2$), $-\frac{d[A]}{dt} \simeq k_1[A]$.

(d) At high concentration of A ($[A] >> k_2$), order of reaction is zero.

3. A bulb of capacity 1 dm^3 contains 1.004×10^{23} gaseous hydrogen molecules and the pressure exerted by these molecules is 101.325 kP_a, the root mean square speed of hydrogen molecules is ($N_A = 6.023 \times 10^{23}$, R = 0.821 L atm mol^{-1} K^{-1} = 8.314 J K^{-1} mol^{-1})

(a) $\sqrt{9.1\times10^5}$ ms^{-1} (b) $\sqrt{3.23\times10^5}$ ms^{-1}

(c) $\sqrt{9.1\times10^4}$ ms^{-1} (d) $\sqrt{3.23\times10^4}$ ms^{-1}

4. Consider the following changes:

1. $M(s) \longrightarrow M(g)$
2. $M(s) \longrightarrow M^{2+}(g) + 2e^-$
3. $M(g) \longrightarrow M^+(g) + e^-$
4. $M^+(g) \longrightarrow M^{2+}(g) + e^-$
5. $M(g) \longrightarrow M^{2+}(g) + 2e^-$

The second ionization energy of M could be calculated from the energy values associated with

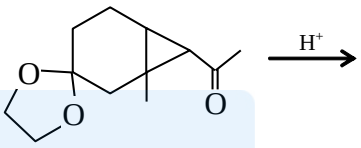
(a) 1 + 5 (b) 2 + 4
(c) 5 – 3 (d) 2 – 1 + 3

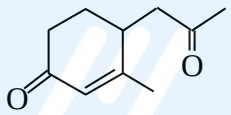
SECTION – II
Multiple Correct Choice Type

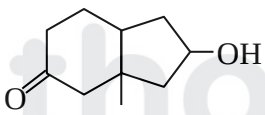
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

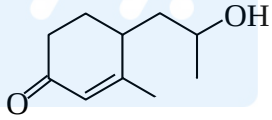
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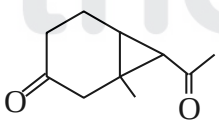
5. Identify the product(s) formed during the following reaction.



(a) 

(b) 

(c) 

(d) 

6. 10 mL of a weak acid (HA) solution is 20% dissociated in water. This solution is completely neutralized by 10 mL of 2×10^{-3} M NaOH solution. Which of the following is/are true about the given case? (log 2 = 0.3)

(a) pH of weak acid (HA) solution was 3.4.
(b) The dissociation constant of weak acid (HA) was 1×10^{-4}.
(c) pH of the solution after addition of NaOH is 7.5.
(d) K_h (hydrolysis constant) of the salt is 1×10^{-9} M.

7. Which of the following statement(s) is/are correct about acetone and trichloromethane mixture?

(a) Mixtures of acetone and trichloromethane show positive deviation from Raoult's law.

(b) The force of attraction acting between molecules of acetone and trichloromethane in a mixture are greater than those acting between the molecules in pure acetone.

(c) Pure acetone can be obtained by the careful fractional distillation of any mixture of acetone and trichloromethane.

(d) When acetone and trichloromethane are mixed, the enthalpy change is negative.

8. For the given reaction, $X_2 + H_2O \longrightarrow X^- + HOX + H^+$

Which of the following statement(s) is/are true?

(a) It is a nucleophilic displacement reaction.

(b) It is a second order reaction.

(c) The unit of rate constant of the reaction is sec^{-1}.

(d) The HOX is a linear molecule.

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True ; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True ; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1:** Aqueous ferrous sulphate is a very good absorber for NO, the compound formed in this reaction is found to contain three unpaired electrons.

Statement-2: Paramagnetic species will always contain odd number of electrons.

(a) A (b) B (c) C (d) D

10. **Statement-1:** Mixture of gases containing one mole each of monoatomic gas and diatomic gas gives value of C_p/C_v as 1.5.

Statement-2: Value of C_p/C_v is independent of moles of the gases taken.

(a) A (b) B (c) C (d) D

11. **Statement-1:** When malonic acid is heated alone, it undergoes decarboxylation whereas heating it with P_2O_5 results in dehydration.

Statement-2: Dicarboxylic acids on heating alone undergo decarboxylation only.

(a) A (b) B (c) C (d) D

12. **Statement-1:** At very high pressure, value of compressibility factor (Z), for all gases is greater than one.

Statement-2: At very high pressure, forces of repulsions dominate over attractive forces.

(a) A (b) B (c) C (d) D

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

On dissolving a sparingly soluble salt in water, after some time an equilibrium is established when the rate of dissolution of ions from the solid equals the rate of precipitation of ions from the saturated solution at a particular temperature. For example, when solid AgCl is added to water, following equilibrium is found to exist.

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$

$$K_{eq} = \frac{[Ag^+][Cl^-]}{[AgCl]}$$

$$K_{sp} = K_{eq}[AgCl] = [Ag^+][Cl^-]$$

$K_{eq}[AgCl]$ is known as solubility product (K_{sp}). The principle of solubility product applies to all solution of slightly soluble materials whether they dissociate in one step or in several steps.

13. The salt $Al(OH)_3$ is involved in the following two equilibria:

$Al(OH)_3(s) \rightleftharpoons Al^{3+}(aq) + 3OH^-(aq)$; K_{sp}

$Al(OH)_3(s) + OH^-(aq) \rightleftharpoons Al(OH)_4^-(aq)$; K_c

Which of the following relationship is correct at the point of minimum solubility of $Al(OH)_3$?

(a) $[OH^-] = \left(\frac{K_{sp}}{K_c}\right)^{1/3}$ (b) $[OH^-] = \left(\frac{K_c}{K_{sp}}\right)^{1/4}$

(c) $[OH^-] = \sqrt{\left(\frac{K_{sp}}{K_c}\right)^{1/4}}$ (d) None of these

14. Consider the following statement and select the correct option.

(I) The concentration of Fe^{3+} will increase in a solution of $Fe(OH)_3$ when $[H^+]$ decrease. (Given: K_{sp} of $Fe(OH)_3$ is 4×10^{-38} at 298 K)

(II) In a mixture of NH_4Cl and NH_4OH in water, a further amount of NH_4Cl is added the pH of mixture will decrease.

(III) An aqueous solution of NH_4I and HCO_2K will be basic and acidic respectively.

(a) only (I) is correct. (b) only (II) is correct.

(c) only (III) is correct. (d) (II) and (III) are correct.

15. The simultaneous solubility of $AgCN(K_{sp} = 2.5 \times 10^{-16})$ and $AgCl$ $(K_{sp} = 1.6 \times 10^{-10})$ in 1M $NH_3(aq)$ are $\{\text{Given: } K_f[Ag(NH_3)_2^+]\} = 10^7$.

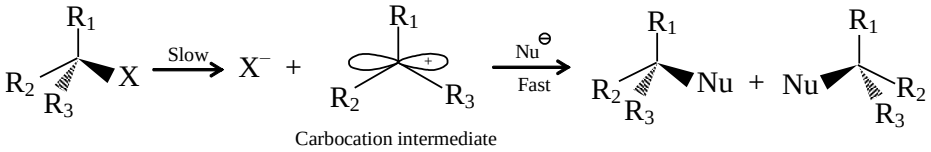
(a) 0.037 & 5.78×10^{-8} respectively (b) 5.78×10^{-8} & 0.037 respectively

(c) 0.04 & 6.25×10^{-8} respectively (d) 1.58×10^{-3} & 1.26×10^{-5} respectively

Passage-II

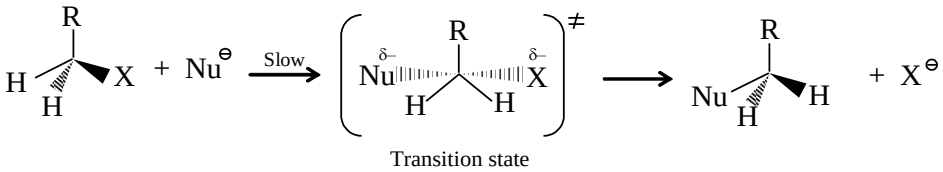
Nucleophilic substitution in alkyl halides usually follows either of the two pathways.

S_N1 process:



This pathway is preferred for tertiary halides.

S_N2 process:



Primary alkyl halides preferably follows S_N2 pathway. Secondary alkyl halides follows dual pathway. Some part of it goes via S_N1 and remaining part of it goes through S_N2 process.

Optically active (S)–(+)–2–bromo octane, $[\propto]_D^{25^o}$ = +36°, reacts with aqueous NaOH in acetone to give optically pure (R)–(–)–2–octanol, $[\propto]_D^{25^o}$ = –10°. With partially racemized bromo compound whose $[\propto]_D^{25^o}$ = +30°, the $[\propto]_D^{25^o}$ of the alcohol product is –6°.

16. The optical purity of partially racemized alcohol produced is approximately
(a) 83% (b) 60% (c) 17% (d) 40%

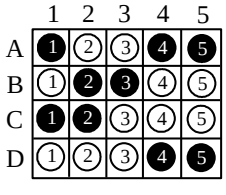
17. The percentage of inversion observed in the reaction is
(a) 66% (b) 28% (c) 72% (d) 34%

18. The percentage of reaction involving front side attack on the bromo compound is
(a) 14% (b) 17% (c) 83% (d) 86%

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

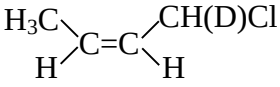
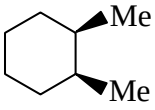
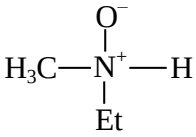
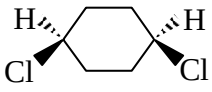
If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

Column I	Column II
(A) $[Cr(gly)_3]$	**1.** Linkage isomerism.
(B) $[CoBr_2Cl_2(SCN)_2]^{4-}$	**2.** Spin only magnetic moment = $\sqrt{15}$.
(C) $[Fe(NH_3)_6]Cl_3$	**3.** Optical isomerism.
(D) $K[PtBrCl(NO_2)(NH_3)]$	**4.** Geometrical isomerism. **5.** Octahedral geometry.

2.

Column I	Column II
(A) 	**1.** Optically active
(B) 	**2.** Optically inactive
(C) 	**3.** Plane of symmetry
(D) 	**4.** Number of chiral center is 1. **5.** Number of stereocenter is 2.

mathongo